

CO4-AT2-Lab Practice

K.Dhilip

192365013

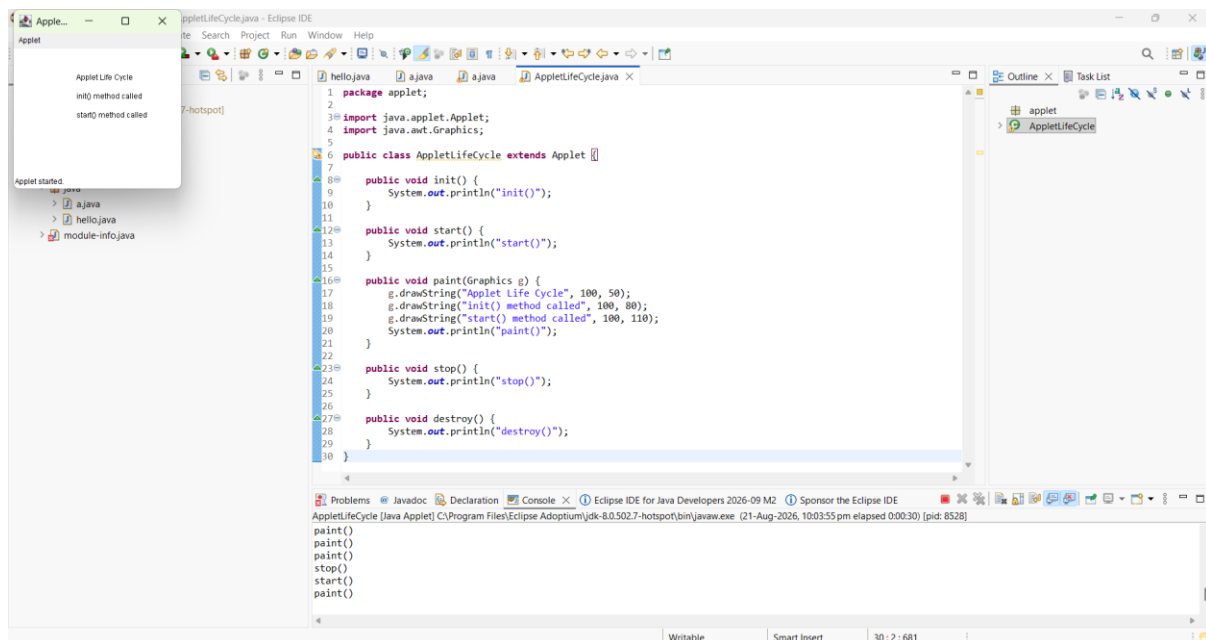
Question 1 - Applet Life Cycle

Develop a Java Applet to demonstrate the Applet Life Cycle methods: init(), start(), paint(), stop(), and destroy().

Display a suitable message in the applet window whenever each life-cycle method is invoked. Observe the sequence of execution by

running the applet and interacting with it Expected Outcome:

Students should understand the execution sequence and purpose of each Applet life-cycle method.



Question 2 - Generating Graphic Images Using Applet

Develop a Java Applet that uses the Graphics class to generate a simple graphical scene.

The applet should draw

A house

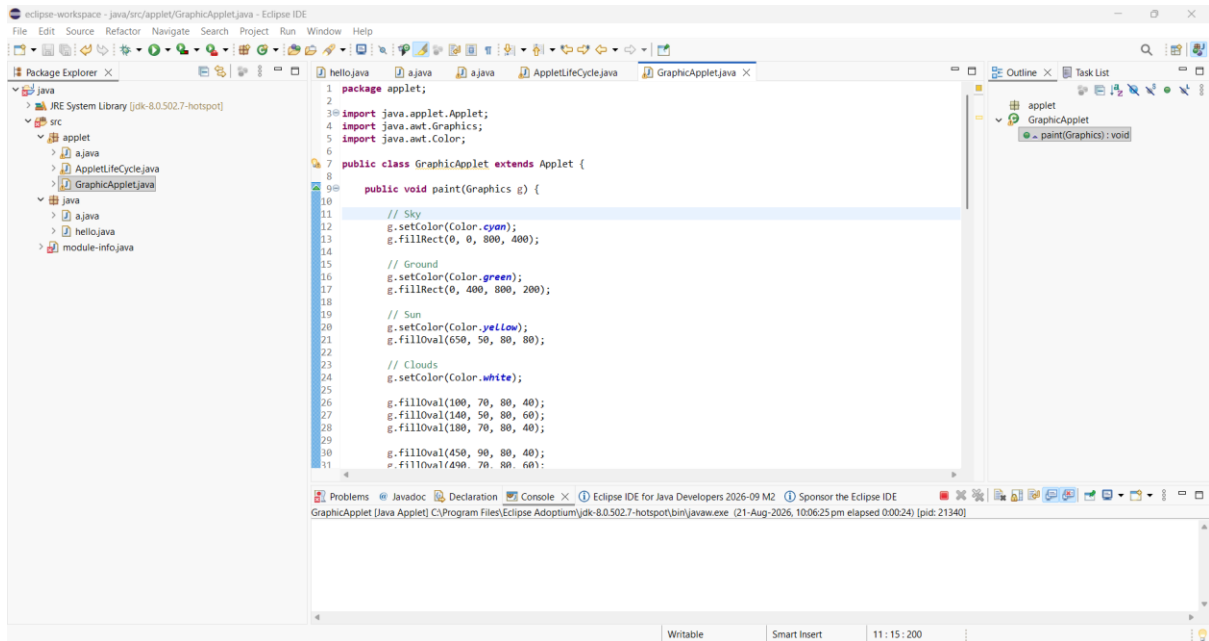
Sun

Trees

Road

Clouds

Use methods such as `drawLine()`, `drawRect()`, `drawOval()`, `fillRect()`, `fillOval()`, and appropriate colors.



Question 3 - Calculator Application Using Applet Parameters

Develop a Calculator Applet that accepts two numbers and an arithmetic operator through HTML/Applet parameters.

The applet should perform

Addition

Subtraction

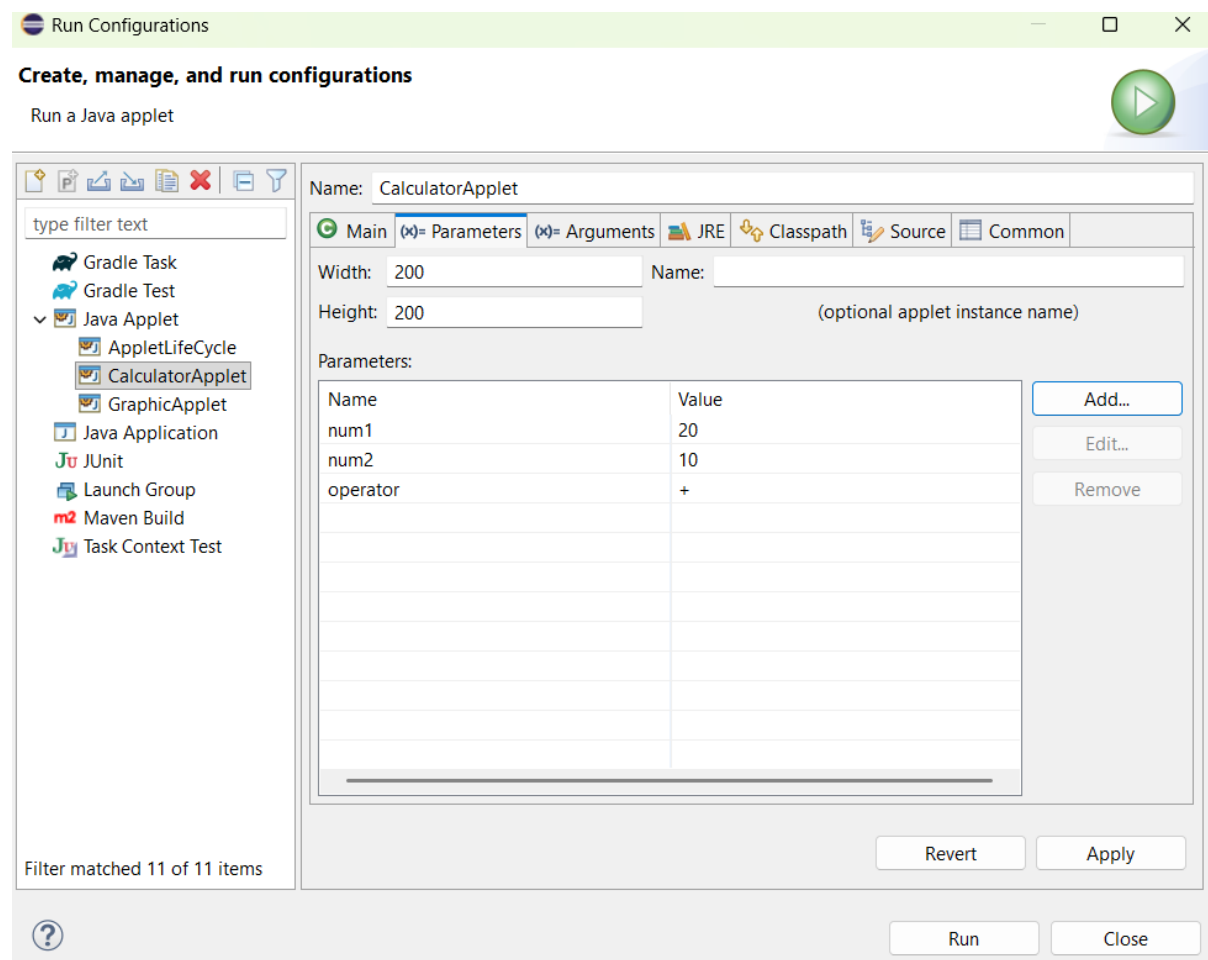
Multiplication

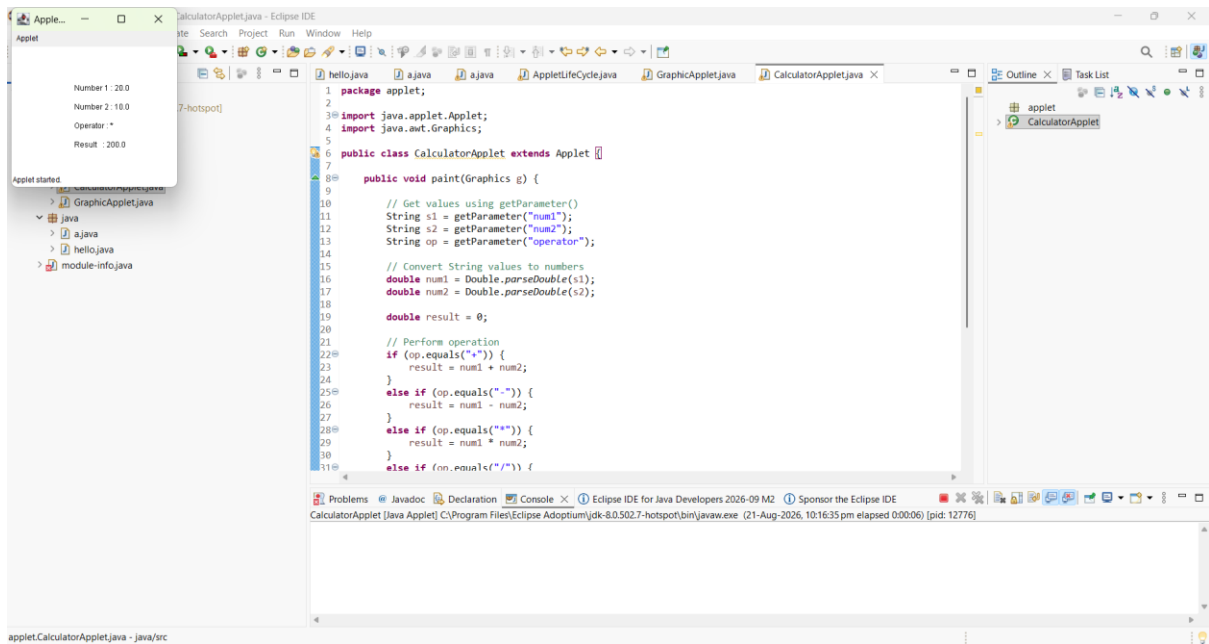
Division

Use the `getParameter()` method to retrieve the values passed to the applet.

Expected Outcome:

Students should understand how parameters are passed from an HTML page to an Applet and processed in Java.





Question 4 - Student Profile Using AWT Components

Develop a Student Profile Application using Java AWT components such as:

Label

TextField

Checkbox

CheckboxGroup

Choice

TextArea

Button

List

The application should accept student details such as:

Name

Register Number

Gender

Department

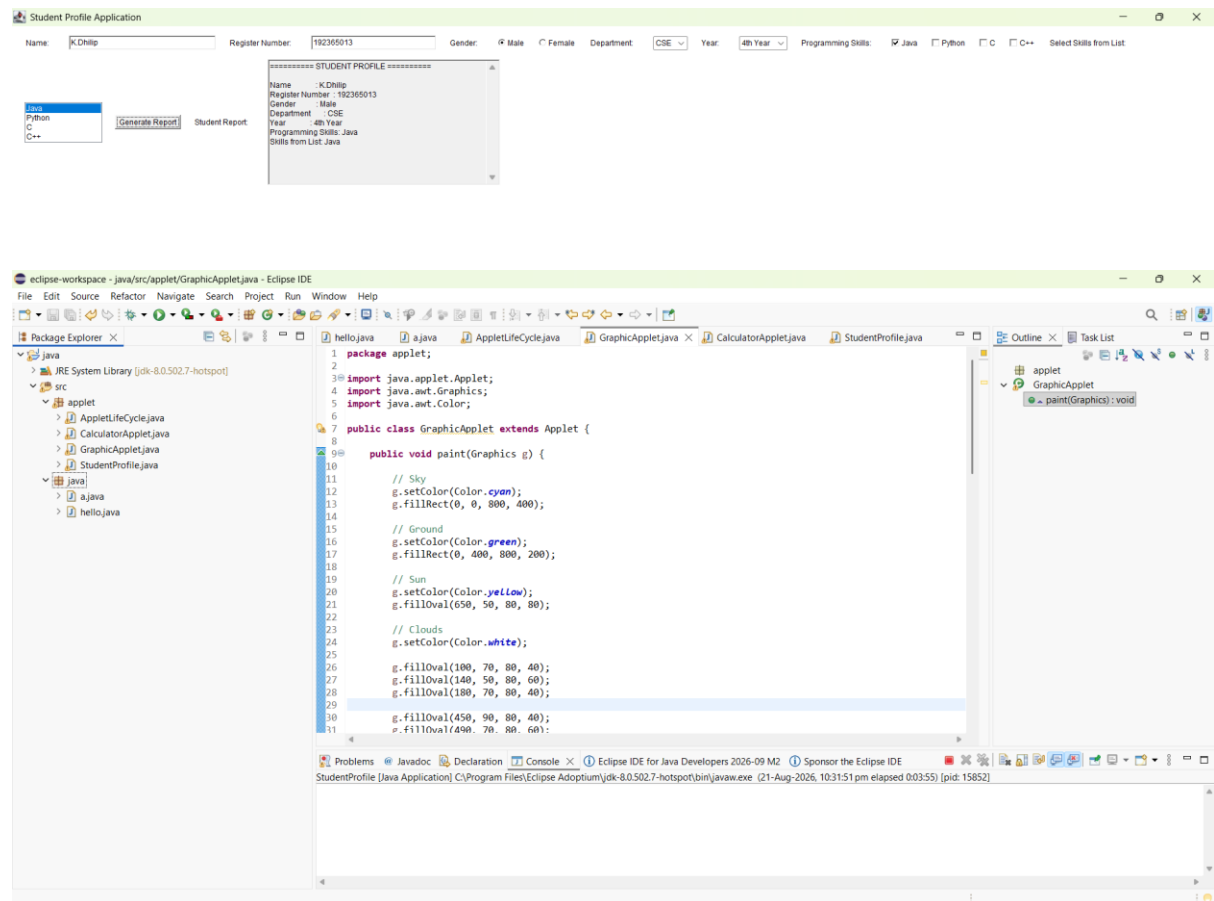
Year

Skills/Programming Languages

When the Generate Report button is clicked, display the entered student information in a List or TextArea.

Expected output:

Students should demonstrate the use of multiple AWT components and event handling,



Question 5 - GridLayout with Event Handling

Develop a Java AWT application using GridLayout to create a simple form or calculator interface. Arrange the components in a grid and implement appropriate event handling using ActionListener. For example, create a calculator with buttons:

+68L

-9Sp

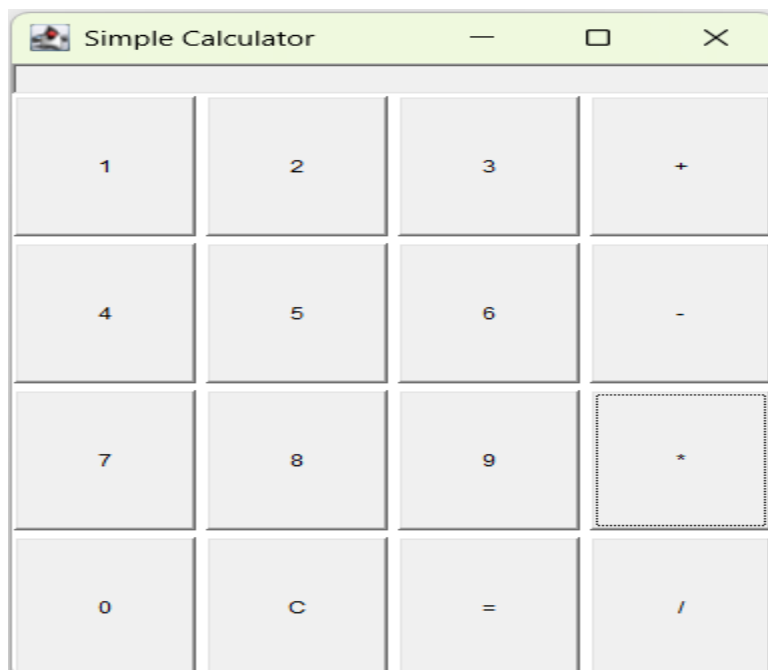
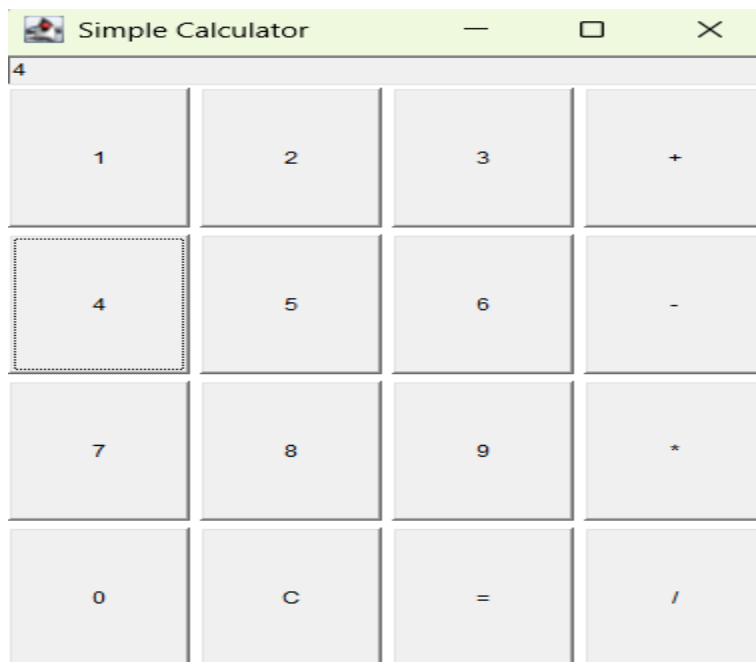
123*

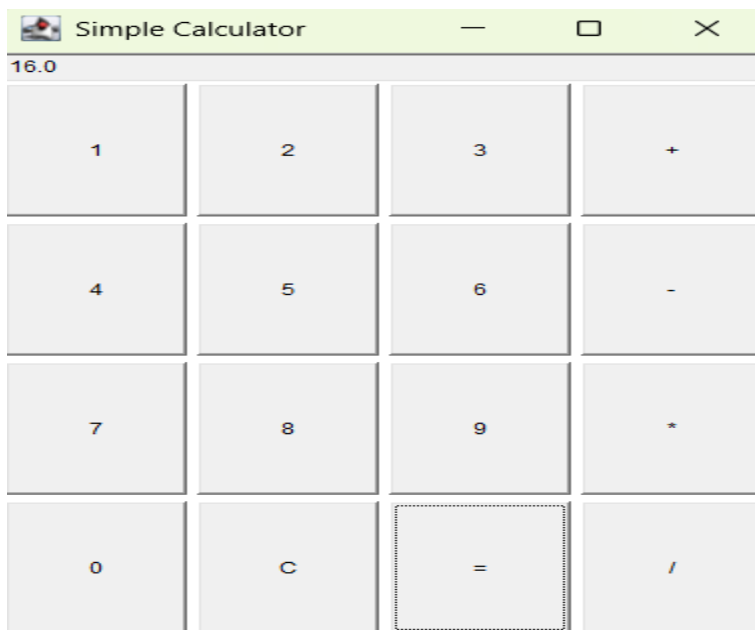
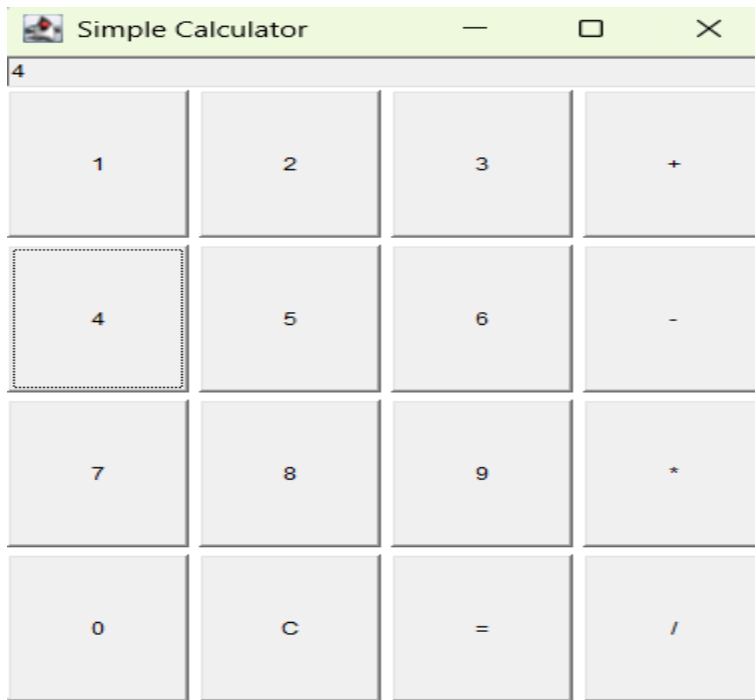
0=C/

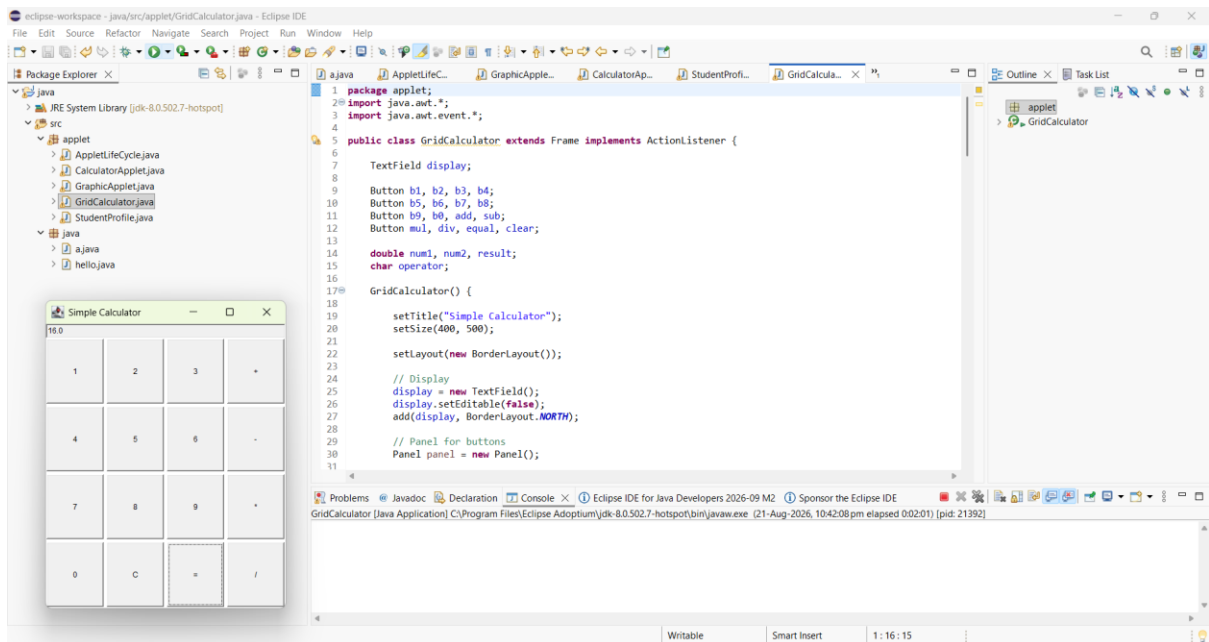
Perform the corresponding operation when a button is clicked.

Expected Outcome:

Students should understand GridLayout and button-based event handling.







Question 6 - GridBagLayout with Event Handling

Develop a Java AWT Student Registration Form using GridBagLayout. The form should contain:

Name

Register Number

Department

Email

Gender

Year

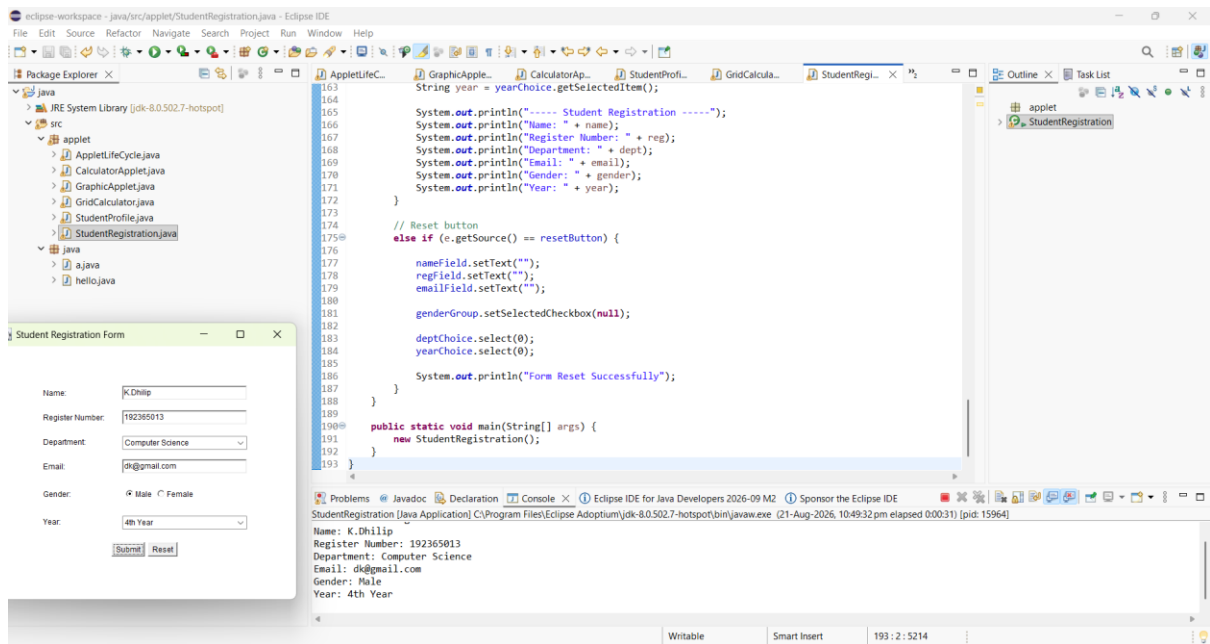
Submit Button

Reset Button

Use GridBagConstraints to properly position and align the components. Implement event handling for the Submit and Reset buttons

Expected Outcome:

Students should understand flexible component positioning using GridBagLayout and event handling.



Question 7 - BorderLayout with Event Handling

Develop a Java AWT application using BorderLayout Divide the window into:

NORTH – Header

SOUTH – Buttons

EAST-Options

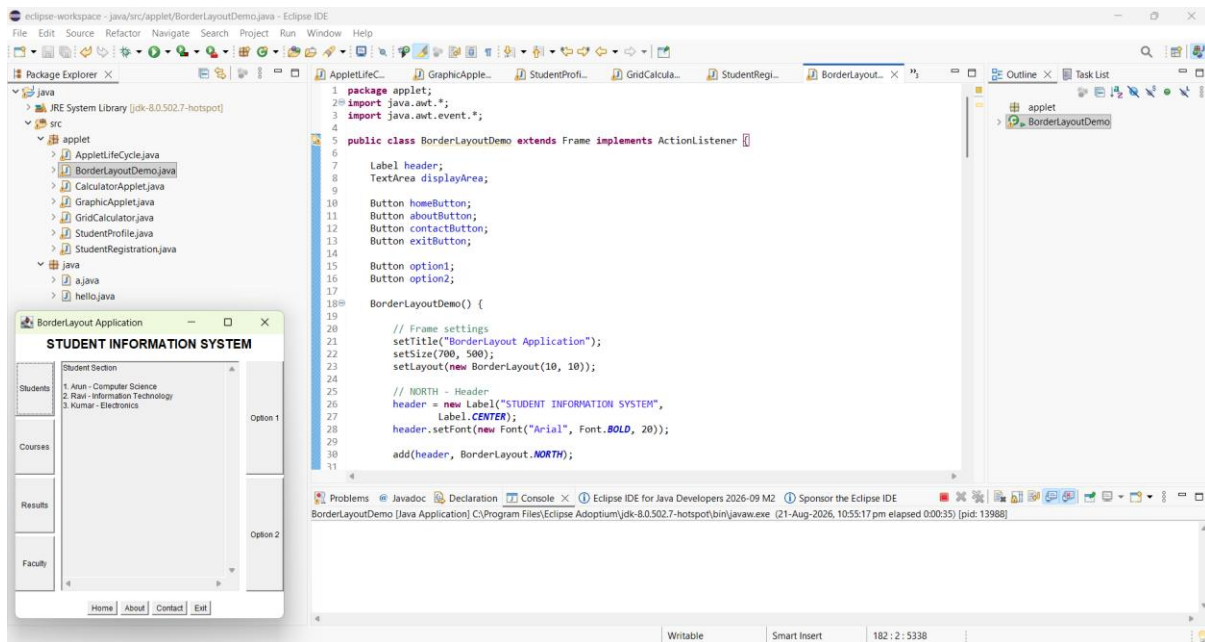
WEST - Menu

CENTER – Display area

Add buttons in the interface and implement event handling so that clicking different buttons changes the content display area

Expected Outcome:

Students should understand the five regions of BorderLayout and event-driven programming



Question 8 - Cardlayout with Event Handling

Develop a Java AWT application using Cardlayout to create a multi-page interface.

Create at least three cards:

Card 1: Student Information

Card 2: Academic Information

Card 3: Confirmation/Report

Provide Next Previous, and Submit buttons to navigate between the cards Implement event handling to switch between cards using CardLayout.

Expected Outcome: Students should understand how CardLayout can be used to create multiple screens within a single window.

